

X-Ray Emitting Plasma in Galaxies and Galaxy Clusters

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Abstract

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1 Introduction

In the study of the evolution of our universe on the largest scales, galaxy clusters represent the most expansive clearly defined structures in which matter approaches virialization. Due to this fact, we can use them as laboratories to study the cosmic growth rate and characterize density fluctuations, as well as reconstruct the dynamical history of merging systems and the effects of feedback from active members. A tenuous plasma called the Intracluster Medium (ICM) permeates these massive objects, with temperatures in the 10^7 K range and densities from 10^{-5} to 10^{-1} cm^{-3} between their outer regions and inner cores. In this hot state, the ICM radiates at several keV photon energies, making spectroscopy in the soft X-ray band a primary method to resolve the distribution of baryonic matter inside clusters (Böhlinger & Werner 2010).

1.1 Cluster Models

At the broadest level, galaxy clusters are modeled as dark matter halos fully governed by gravitational collapse. They are thought to originate from primordial density fluctuations and grow by accreting matter from cosmic filaments as well as from cluster mergers. The overall mass is dominated by the dark component defined through its purely gravitational interactions, the density of which is well described using a Navarro-Frenk-White (NFW) profile.

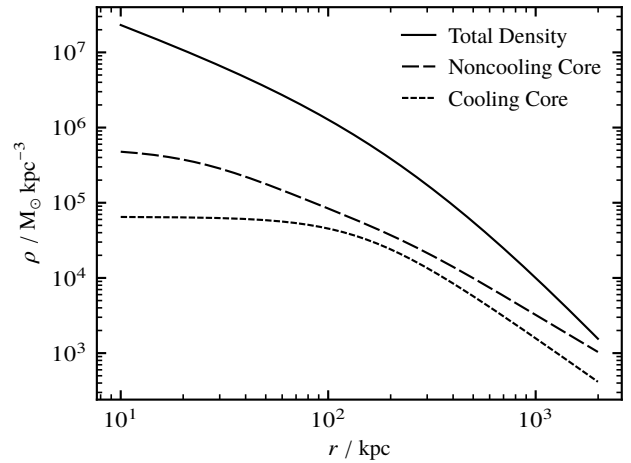


Figure 1: Typical mass density profiles for NCC and CC galaxy clusters are plotted as proxies for all baryonic matter. Shown as well is a standard NFW total density curve that is dominated by a collisionless dark matter component.

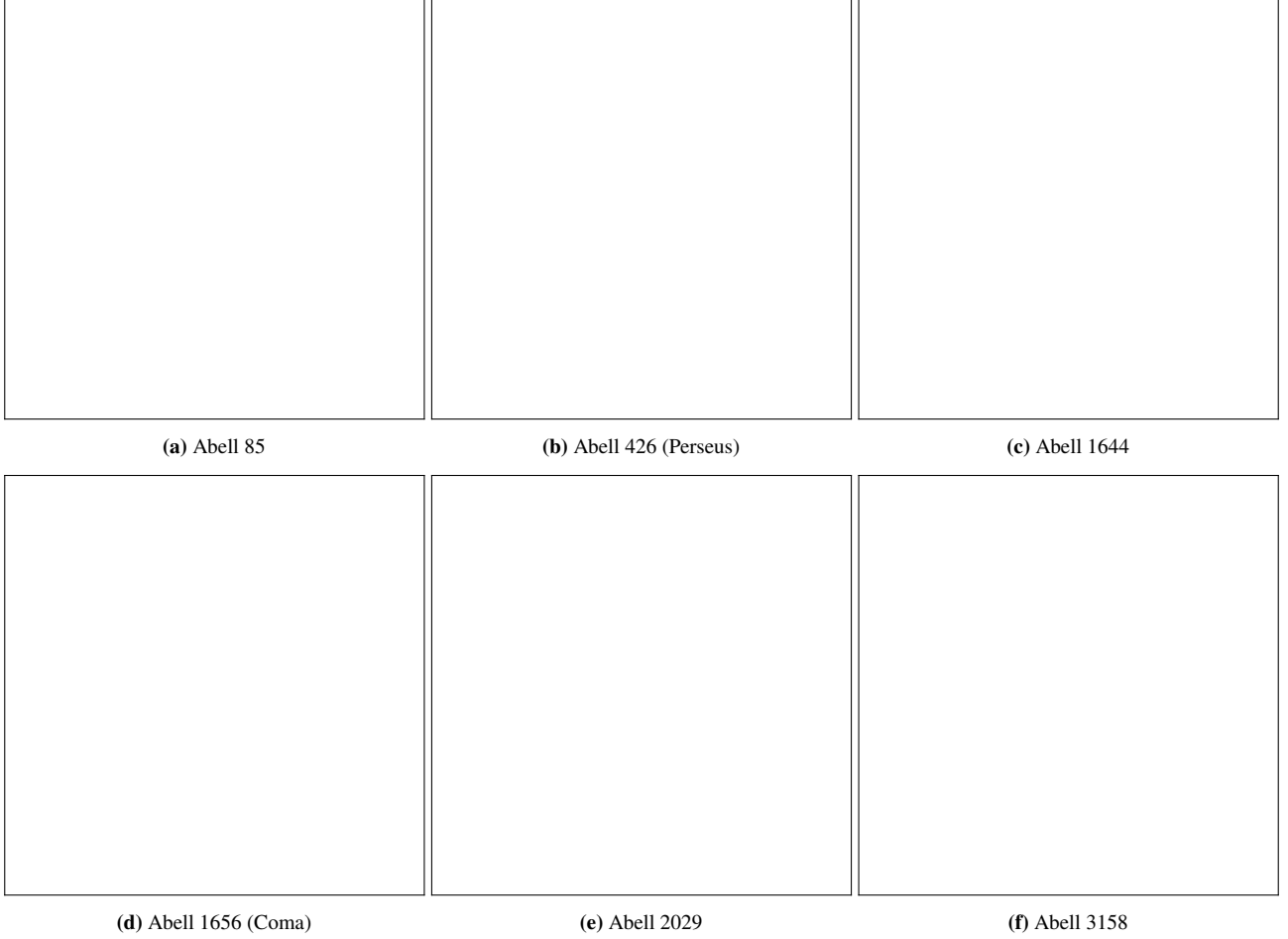


Figure 2: Selection of galaxy clusters used as examples in this work. Displayed here are composite overlays of second generation *Digitized Sky Survey* red band images and contours for *ROSAT All Sky Survey (RASS)* broad band count rates (Truemper 1982, Lasker 1994). Both optical and X-ray data are stretched using a simple 0.8 exponent power law. Each frame has a radius of $15 \times 10^0'$ for its field of view. Refer to **Figure 8** for raw *ROSAT* counts underlying the contours.

2 Radiation Mechanisms

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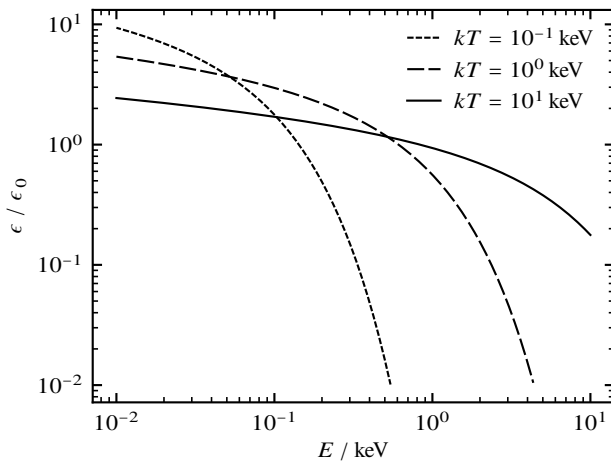


Figure 3: Normalized bremsstrahlung continua for different temperatures over a range of magnitudes. Fitting the bend placement allows for the inference of local temperature.

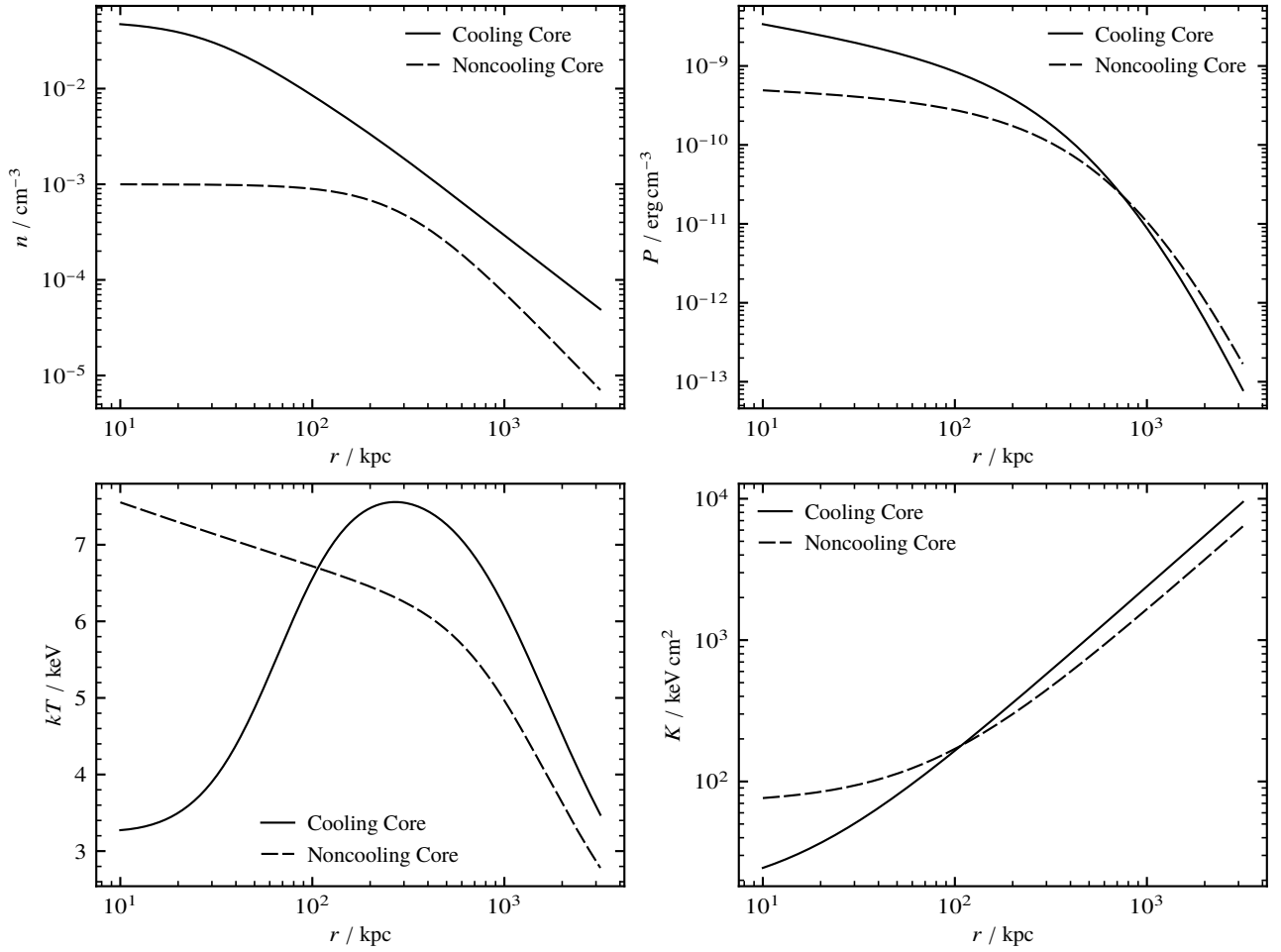


Figure 4: Theoretical galaxy cluster radial profiles assuming universal selfsimilar scaling relations. *Top Left:* Electron number density (Vikhlinin et al. 2006). *Top Right:* Gas pressure (Arnaud et al. 2010). *Bottom Left:* Gas temperature (Vikhlinin et al. 2006). *Bottom Right:* Gas entropy (Cavagnolo et al. 2009). Compare to **Figure 7** for real observations.

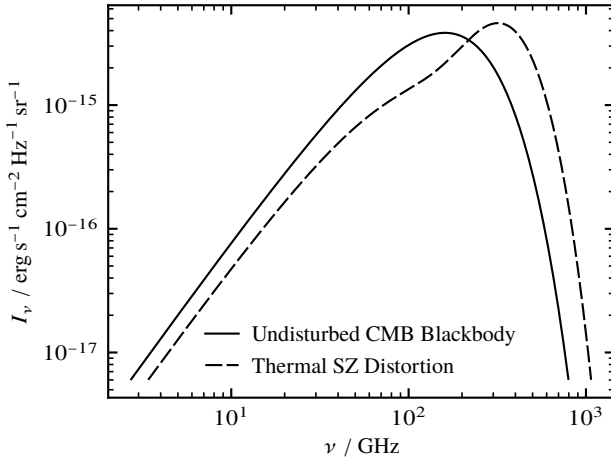


Figure 5: Schematic view of the thermal Sunyaev-Zeldovich effect on the Cosmic Microwave Background (CMB) Planck spectrum with exaggerated $Y \sim 0.2$ Compton parameter in the nonrelativistic limit.

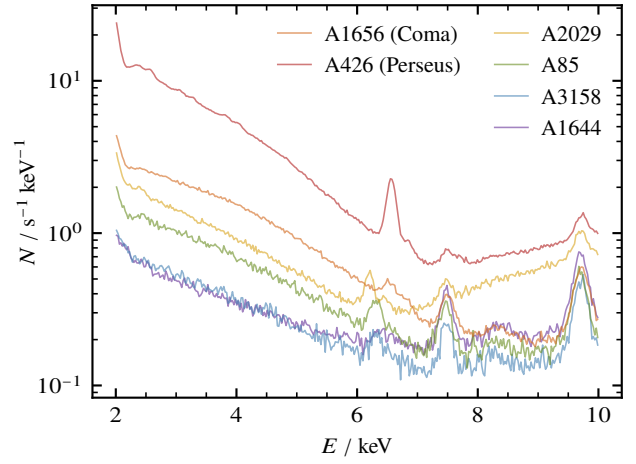


Figure 6: Collection of *Chandra* binned spectra for various galaxy clusters (Weisskopf et al. 2000). Note the presence of emission lines, as well as possible hints of excesses in the relative count rates for both the soft and hard X-ray regimes.

3 Instrumentation

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4 Spectral Characteristics

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5 Plasma Diagnostics

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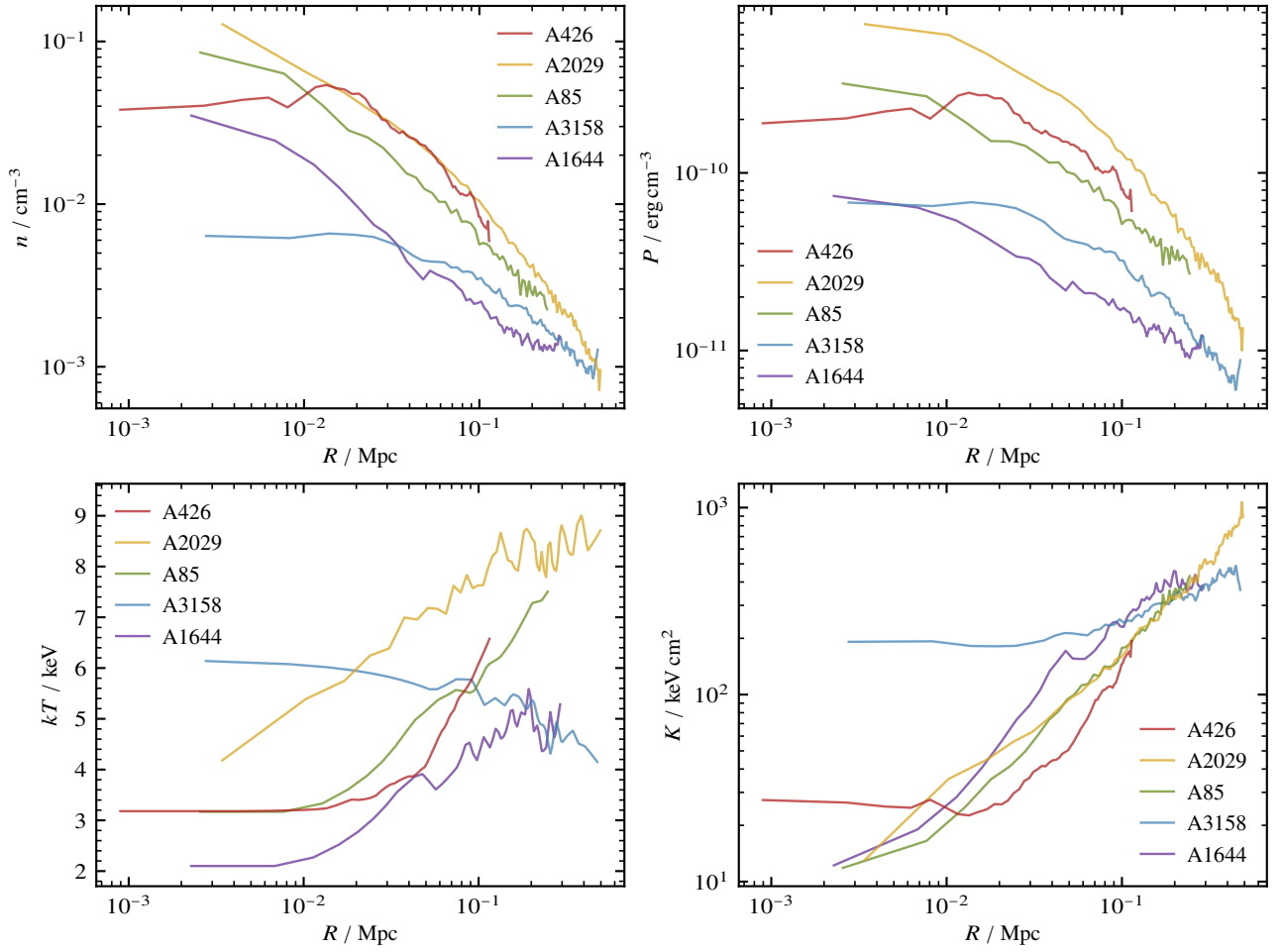


Figure 7: Sample of galaxy cluster radial profiles from *Chandra* observations (Weisskopf et al. 2000). *Top Left:* Electron number density. *Top Right:* Gas pressure. *Bottom Left:* Gas temperature. *Bottom Right:* Gas entropy. Compare to **Figure 4** for a visualization of theoretical and empirical fits with realistic parameter choices.

6 Conclusion

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References

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Böhringer, H. & Werner, N. *The Astronomy and Astrophysics Review* (2010)
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Weisskopf, M. C. et al. *Proceedings of the Society of Photo Optical Instrumentation Engineers* (2000)

Appendix

A Reference Images

B Cluster Profiles

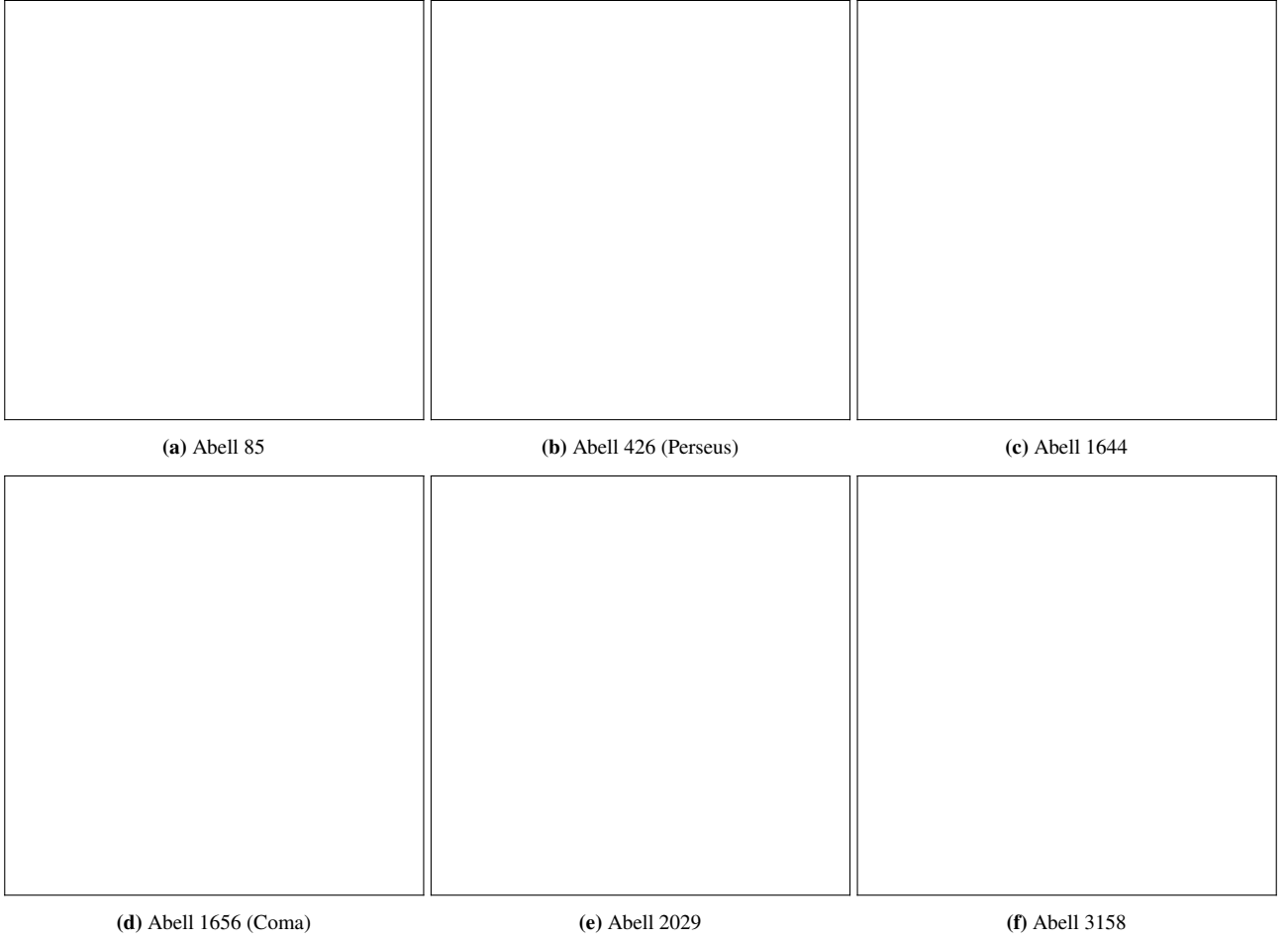


Figure 8: Depicted are *RASS* counts used for X-ray contours shown in **Figure 2** after applying a stretch and Gaussian blur (Truemper 1982). The sampled clusters are selected based on brightness and morphology, as well as data accessibility in the *Archive of Chandra Cluster Entropy Profile Tables (ACCEPT)* and *XMM Cluster Outskirts Project (X-COP)* catalogs (Cavagnolo et al. 2009, Eckert et al. 2017).

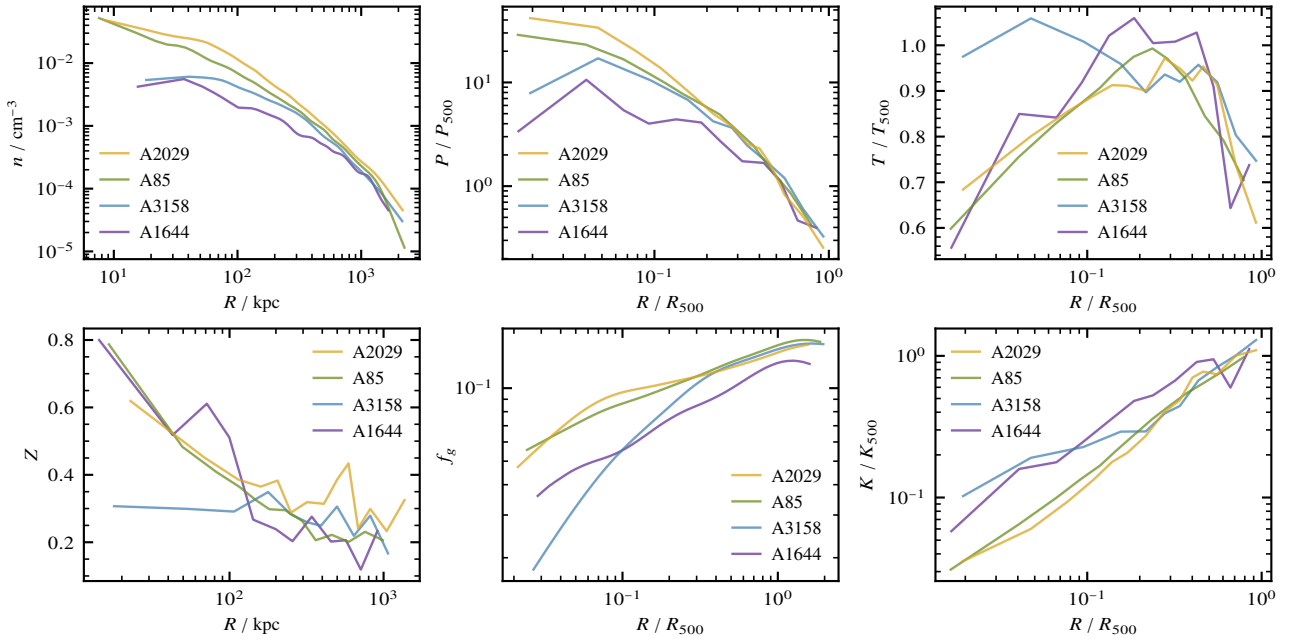


Figure 9: Plots of extracted radial profile projections for available *X-COP* galaxy clusters. *Upper Left:* Electron number density. *Upper Middle:* Gas pressure. *Upper Right:* Gas temperature. *Lower Right:* Gas entropy. *Lower Middle:* Gas mass fraction. *Lower Left:* Metal abundance. Consult the text for definitions of characteristic quantities and scaling. Compare as well to reconstructed distributions in **Figure 7** which focus on the inner regions instead of the cluster outskirts.